MIL-C-83295A(USAF)
5 April 1972
SUPERSEDING
MIL-C-83295 (USAF)
18 January 1971

MILITARY SPECIFICATION

COT, FOLDING, DOUBLE-DECKING MXU-615/E82; LIGHTWEIGHT, BARE BASE

1. SCOPE

1.1 This specification covers one type of lightweight, double-decking folding cot, designated MXU-615/E82.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

QQ-P-416	Plating, Cadmium (Electrodeposited)
DDD-L-20	Label, for Clothing, Equipage, and Tentage, (General Use)

* Military

MIL-P-116	Preservation, Methods of
MIL-A-8625	Anodic Coatings, for Aluminum and Aluminum Alloys

STANDARDS

Federal

FED-STD-751	Stitch, Seams and Stitching

* Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-130	Identification Marking of U. S. Military Property
MIL-STD-794	Parts and Equipment, Procedures for Packaging and Packing of
MIL-STD-831	Test Reports, Preparation of

FSC 7105

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DRAWINGS

Army Natick Laboratories

2-5-79	Cot, Folding, Aluminum, Double-Decking; Assembly Complete
2-5-80	Cot, Folding, Aluminum, Double-Decking; Cover Assembly
2-5-81	Cot, Folding, Aluminum, Double-Decking; Frame Assembly
2-5-82	Cot, Folding, Aluminum, Double-Decking; Leg Assembly

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Preproduction. This specification makes provisions for preproduction testing.

3.2 Materials and components. Materials and components shall be as specified herein.

3.3 Design and construction. The cot shall conform to the design and construction requirements shown on Drawings 2-5-79, 2-5-80, 2-5-81, and 2-5-82 with the exceptions and additions specified herein.

3.3.1 Machine stitching. All stitching, except bartacking, shall conform to type 301 of FED-STD-751, 6 to 8 stitches per inch.

3.3.1.1 Thread breaks. Thread breaks in stitching shall be overstitched not less than 1 inch at each break. Bartacks shall be free of thread breaks.

3.3.1.2 Stitching ends. Ends of stitching shall be overstitched not less than 1 inch, except where ends are turned under in a hem, held down by other stitching, or where stitching is performed automatically. When stitching is performed automatically, three or more tying, overlapping, or backstitches shall be used to secure the ends.

3.3.1.3 Skipped stitches. Two or more consecutively skipped stitches shall be overstitched not less than 1 inch.

3.3.2 Feet. Plastic feet shall be cemented into the legs of the cot. The adhesive shall be such as recommended by the adhesive manufacturer for adherence of plastic to aluminum.

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3.3.3 Setting of grommets. Holes punched to receive grommets shall be smaller than the outside diameter of the grommet barrel so that the barrel must be forced through the hole. Grommets shall be securely clinched without cutting the materials.

3.4 Finish of frame. Prior to fabrication of the cot, aluminum tubing shall be anodized in accordance with type II, class 1 of MIL-A-8625. Steel saddles shall be given a cadmium plate conforming to class 3, type I of QQ-P-416.

3.5 Markings

3.5.1 Frame. The letters US, the name of the supplier, and the number and date of the contract shall be stamped on the frame of the cot in accordance with MIL-STD-130.

3.5.2 Cover. Identification marking on the cover shall conform to type IV, class 5 of DDD-L-20 and shall be located as shown on Drawing 2-5-80.

3.6 Assembly instructions. The following instructions for assembling the cot shall be legibly printed in indelible ink on a sheet of waterproof paper a minimum of 10 by 15 inches, or shall be stenciled on the underside of the fabric.

INSTRUCTIONS FOR ASSEMBLING COT

Step 1. Insert tubing of one "U" frame into sewn sleeves of nylon cover. Gather cover completely onto frame, to expose ends of frame tubing.

Step 2. Connect second "U" to frame with cover by inserting ends of tubing into the respective connecting sleeve.

Step 3. Evenly spread out nylon cover over assembled "U" frames and firmly lace the cover to the frames by means of the lacing cords and flaps. Secure ends of cord.

Step 4. Depress leg-locking buttons and pivot legs outward until locked in open position.

For double-decking of cots, procede with the following steps:

Step 5. Assemble two cots as indicated in steps 1 through 4. Next, extend the four outer, or telescoping, legs of one cot by depressing the locking buttons located near the bottom of the legs and extending the legs until locked in position.

Step 6. Position the cot with extended legs onto the second cot and secure in place with the pins and springs which are attached to the saddles of the lower cot.

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3.7 Workmanship. Metal components of the cot shall be free of burrs, sharp edges, and corroded areas, and shall not be broken or malformed. Rivets shall be set and neatly headed. Cloth components shall be clean and free of holes, cuts, or tears, or cloth defects such as multiple floats, broken or missing yarns. Tape shall contain no frayed or scalloped edges. Thread tension shall be maintained so that there will be no loose stitching; and seam allowances shall be maintained with seams properly sewed so that no runoffs, pleats, or open seams will result. Care shall be taken in sewing so that there will be no needle chews. Thread ends shall be trimmed to 1/2 inch or less. Cord components shall be free of cuts or abrasions.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

* 4.2 Classification of inspections. The inspection requirements shall be classified as follows:

a. Preproduction inspection 4.3

b. Acceptance inspection . . . . . 4.4.

*

* 4.3 Preproduction inspection (see 6.2)

* 4.3.1 Inspection sample. One cot shall be subjected to the examinations specified in 4.3.3.

* 4.3.2 Test report. After completion of the preproduction inspections, a test report shall be prepared in accordance with MIL-STD-831 and three complete copies of the report furnished the procuring activity.

* 4.3.3 Preproduction inspection. The preproduction inspection shall consist of the examination specified in 4.4.1.1 and 4.4.1.2.

* 4.4 Acceptance inspection. The acceptance inspection shall consist of the sampling inspection.

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* 4.4.1 Sampling inspection. Sampling for inspection shall be performed in accordance with MIL-STD-105. The lot shall consist of all complete cots offered for inspection at one time. The samples shall be subjected to the following:

* 4.4.1.1 Visual examination. Visual examination shall be made of the cot for the defects in table I. The inspection level shall be level II and the acceptable quality levels (AQL) shall be 6.5 major, 40.0 total (major and minor defects combined).

* 4.4.1.2 Dimensional examination. Dimensional examination shall be made of the completely fabricated cot. Any dimension not within the specified requirements shall be classified as a defect. The inspection level shall be level II, the AQL shall be 6.5 major, 40.0 total (major and minor defects combined).

* 4.5 Inspection of preparation for delivery. Preservation, packaging, packing, and marking shall be inspected to determine conformance to the requirements of section 5 herein.

5. PREPARATION FOR DELIVERY

* 5.1 Packaging

* 5.1.1 Level A or C. The cot shall be packaged in accordance with MIL-STD-794, level A or C, as specified (see 6.2). The preservation method shall be selected from the appendix of MIL-STD-794 and applied in accordance with MIL-P-116.

* 5.2 Packing. Packing shall conform to MIL-STD-794, level A, B, or C, as specified (see 6.2).

* 5.3 Marking. The exterior and interior containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The MXU-615/E82 cot is intended for use by personnel engaged in BARE BASE operations. The cot is designed for use in both individual cot arrangements or double-decked to temporarily accept higher than normal occupancy levels within a shelter caused by any loss of facilities or other need which may require doubling-up.

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TABLE I. Classification Of Defects For Visual Characterisitcs

Examine	Defect	Major	Minor
Cloth	Hole, cut, or tear	X	
	Any smashed, broken, or missing yarn; multiple floats, clearly visible at normal inspection distance (approximately 3 feet)	X	
Tape, binding	Edge frayed or scalloped		X
Cord, braided	Any cut	X	
	Any abrasion		X
	Free ends not finished as specified		X
Seams and stitching	Open seam - 1/2 inch or less		X
	- more than 1/2 inch	X	
	NOTE: A seam shall be classified as open when one or more stitches joining a seam or broken or when two or more consecutively skipped stitches or runoff stitches occur. On double-stitched seams, a seam shall be classified as open when either one or both sides of a seam are open.		
	Raw edges caught in stitching (where turnunder is required) - more than 1 inch, but not more than 3 inches		X
	- more than 3 inches	X	
	Thread breaks overstitched less than 1 inch		X
	Skipped stitches overstitched less than 1 inch (where two or more consecutively skipped stitches occur)		X
	Needle chews resulting in cut, tear, or hole	X	
	One or more required rows of stitching omitted	X	

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TABLE I. Classification Of Defects For Visual Characteristics (Cont'd)

Examine	Defect	Major	Minor
Seam type	Wrong seam type	X	
Stitch type	Wrong stitch type	X	
Stitching gage	Not as specified		X
Stitching margin	Larger than specified by more than 1/8 inch: - for more than 1 inch but not more than 4 inches in length		X
	- for more than 4 inches in length	X	
Stitch tension	Loose, resulting in an exposed bobbin or top thread for more than 6 inches in length		X
	Tight, as evidenced by puckering on seams for more than 6 inches in length	X	
Stitching ends	Overstitched less than 1 inch (except where ends are held down by other stitching, turned under in a hem, or where stitching is performed automatically)		X
	Secured with less than three tying, overlapping, or backstitches when automatic stitching is performed		X
Stitches per inch	Less than minimum specified: - one stitch		X
	- two or more stitches	X	
	NOTE: Variation in the number of stitches per inch, caused by the operator speeding up the machine and pulling the fabric in order to sew over heavy places or heavy seams or in turning corners, shall be classified as follows:		
	a. Within the minor defect classification - no defect		
	b. Within the major defect classification - minor defect		

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TABLE I. Classification Of Defects For Visual Characteristics (Cont'd)

Examine	Defect	Major	Minor
Grommets	Improperly set		X
	Clinched loosely, allowing grommet to rotate around hole		X
	Clinched excessively tight; ie, cutting cloth	X	
	Insecurely clinched, ie, grommet or washer becomes disengaged	X	
	Burr or sharp edge		X
Frame components			
General	Any component broken	X	
	Malformed or bent: - fails to perform intended function	X	
	- but will perform intended function		X
	Dented		X
	Burr, sharp edge	X	
Rivets	Not set or neatly headed	X	
Nuts	Missing or threads stripped	X	
Plastic feet	Not adhered to leg or fastened in place		X
	Missing	X	
Marking	Omitted, incorrect, illegible, or misplaced		X
Cleaness	Grease or oil stains, thread ends not trimmed to 1/2 inch or less throughout		X

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6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification

b. Location and conditions for preproduction inspection (see 4.3)

* c. Applicable levels of preservation, packaging, and packing (see 5.1 and 5.2).

* 6.3 Identification of changes. The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This has been done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this documents based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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Reviewer:
Air Force - 84

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NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

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